

Depression, Study Effort, and Productivity: An Empirical and Dynamic Programming Analysis

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1 Introduction

Depression is known to affect motivation, concentration, and overall academic performance. In this project, I study how depression relates to students' study behavior and GPA outcomes using a dataset of 27,901 university students. I first document basic differences between depressed and non-depressed students in study hours, sleep, and financial stress. I then estimate a GPA production function that shows depressed students have lower marginal productivity of study time.

To better understand how these productivity differences translate into study choices, I build a simple dynamic programming model in which students choose daily study hours to balance GPA gains against effort costs. Using parameters calibrated from the data, the model predicts that depressed students optimally choose to study more hours, despite being less productive. This project combines empirical analysis with a behavioral model to explain why effort patterns differ across depression status.

2 Data and Cleaning

The analysis uses the *Student Depression Dataset* (Zubair Dhuddi, 2023), a publicly available Kaggle dataset containing self-reported information on demographics, study habits, mental health indicators, and academic outcomes for 27,901 university students across multiple

Indian cities. Because the dataset represents only Indian students, external validity should be interpreted within this geographic and institutional context.

2.1 Key Variables

The core variables used in estimation are:

- $CGPA_i$: cumulative grade point average (0–10 scale).
- $Hours_i$: daily study hours (cleaned from the “Work/Study Hours” field).
- $Dep_i \in \{0, 1\}$: indicator for self-reported depression.
- $Sleep_clean$: numeric sleep duration (cleaned from text responses).
- Demographic controls: Age, Gender, Degree major.

2.2 Cleaning Procedures

The raw dataset contains heterogeneous text formats and several fields requiring normalization. All cleaning procedures are implemented programmatically in Python to ensure full reproducibility. The main steps are:

1. **Sleep Duration:** Converted to numeric hours using lightweight text parsing (e.g., midpoints of ranges, `hh:mm` conversion, numeric extraction, and recognition of simple spelled-out numbers).
2. **Study Hours:** The “Work/Study Hours” field is standardized into a numeric variable. Ranges, `hh:mm` formats, and standalone numbers all parse successfully for the full sample.
3. **Financial Stress:** Non-numeric tokens (e.g., “?”, “NA”) are mapped to missing values. For the remaining entries, the first numeric value is extracted to construct a continuous measure.

4. **Binary Variables:** Yes/No fields (suicidal thoughts, family mental illness, depression) are standardized into binary indicators using a robust parser that handles common spelling variants and punctuation.

Quality-assurance checks confirm that:

- Study Hours convert successfully for all 27,901 observations,
- 27,898 observations contain valid numeric Financial Stress values (the remainder consist primarily of non-informative entries).

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3 Descriptive Results

Table 1 presents summary statistics for the main variables by depression status. Of the 27,901 students in the sample, 16,336 (58.5%) report experiencing depression, while 11,565 do not. Mean CGPA is similar across groups (7.62 vs. 7.68), with depressed students exhibiting a slightly higher average. More notable differences appear in study behavior and financial well-being: depressed students report studying approximately 1.6 more hours per day and exhibit substantially higher financial stress. Sleep duration is lower among depressed students, consistent with patterns often associated with academic pressure and mental health concerns.

¹Data sourced from Zubair Dhuddi (2023), *Student Mental Health & Depression Indicators Data*, Kaggle. Available at: <https://www.kaggle.com/datasets/zubairdhuddi/student-dataset>.

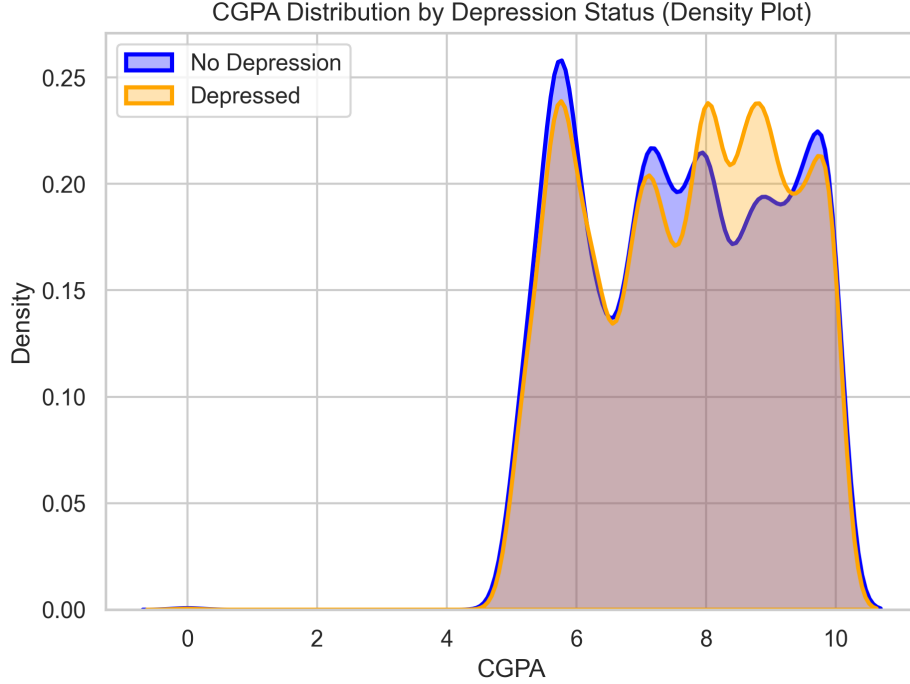


Figure 1: CGPA distribution by depression status (density plot). Depressed and non-depressed students have similar CGPA distributions; the small mean difference is visible but modest relative to within-group spread.

	Non-Depressed	Depressed
Number of students	11,565	16,336
Mean CGPA	7.62	7.68
Mean study hours	6.24	7.81
Mean sleep hours	6.40	6.18
Mean financial stress	2.52	3.58

Table 1: Descriptive statistics by depression status.

To formally assess these differences, we conduct Welch two-sample t -tests for CGPA, study hours, and sleep duration. All three variables differ significantly across depression status. Depressed students study, on average, 1.57 additional hours per day ($p < 0.001$), sleep 0.22 fewer hours ($p < 0.001$), and have a small but statistically significant higher CGPA (difference of 0.066, $p < 0.01$). These differences suggest meaningful behavioral and environmental distinctions between the two groups.

Figure 2 visualizes the distribution of reported study hours by depression status. De-

pressed students not only study more on average but also display considerably greater dispersion: the upper tail of the distribution extends beyond 10–12 hours, while non-depressed students have a more compressed range. This pattern aligns with the elevated financial stress reported by depressed students and may reflect compensatory study behavior or greater academic pressure.

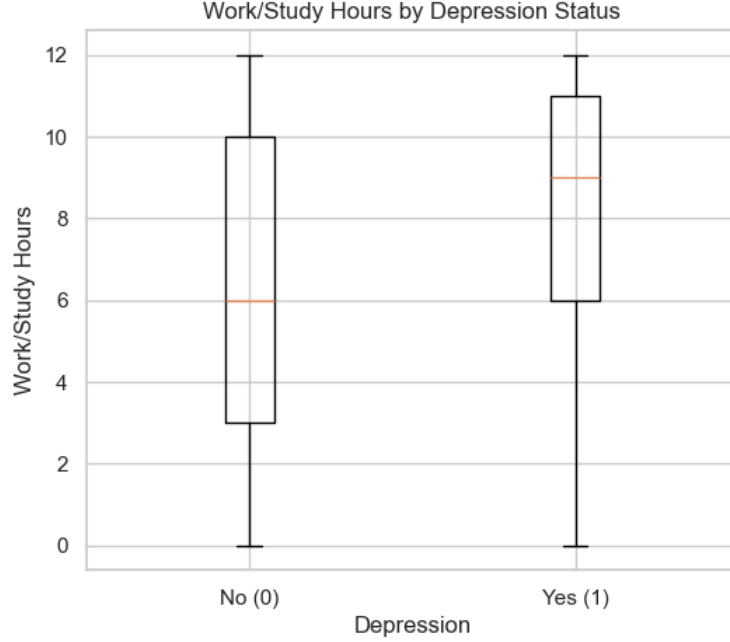


Figure 2: Distribution of reported study hours by depression status. Depressed students report both higher average study hours and greater dispersion.

4 GPA Production Function

We model academic performance as the outcome of a simple production function in which study hours contribute to GPA with diminishing returns. Let H_i denote daily study hours for student i and $Dep_i \in \{0, 1\}$ indicate depression. The GPA production function is specified as:

$$CGPA_i = \beta_0 + \beta_1 H_i + \beta_2 H_i^2 + \gamma_1 Dep_i + \gamma_2 (Dep_i \cdot H_i) + \gamma_3 (Dep_i \cdot H_i^2) + u_i,$$

where $\beta_1 > 0$ and $\beta_2 < 0$ capture diminishing marginal returns to study effort. The interaction terms allow both the level and slope of the production function to differ for depressed students.

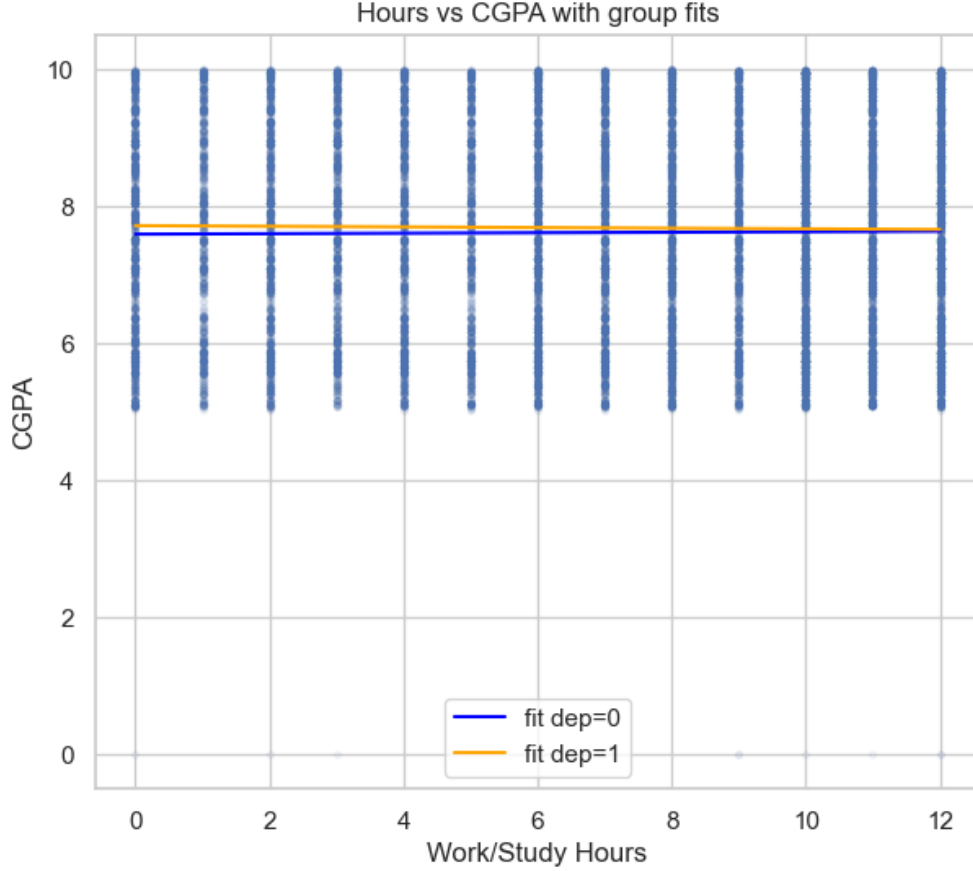


Figure 3: Scatter plot of study hours and CGPA with group-specific fitted lines. The fitted line for depressed students is flatter, indicating lower marginal returns to study hours.

5 Marginal Productivity of Study Time

The marginal product of an additional hour of study is:

$$MP_i = \frac{\partial CGPA_i}{\partial H_i} = \begin{cases} \beta_1 + 2\beta_2 H_i, & \text{if } Dep_i = 0, \\ (\beta_1 + \gamma_2) + 2(\beta_2 + \gamma_3) H_i, & \text{if } Dep_i = 1. \end{cases}$$

Thus, depression changes both the overall GPA level and the effectiveness of additional study hours. These marginal products are used later in the dynamic programming analysis to determine optimal study-hour choices

Using the estimated coefficients from the interaction regression, the marginal products evaluated at the mean study hours of each group are:

$$MP_{\text{non-dep}}(H = 6.238) = 0.003239, \quad MP_{\text{dep}}(H = 7.808) = 0.002106.$$

A convenient measure of the productivity gap is the ratio

$$e = \frac{MP_{\text{dep}}}{MP_{\text{non-dep}}} \approx \frac{0.002106}{0.003239} \approx 0.65.$$

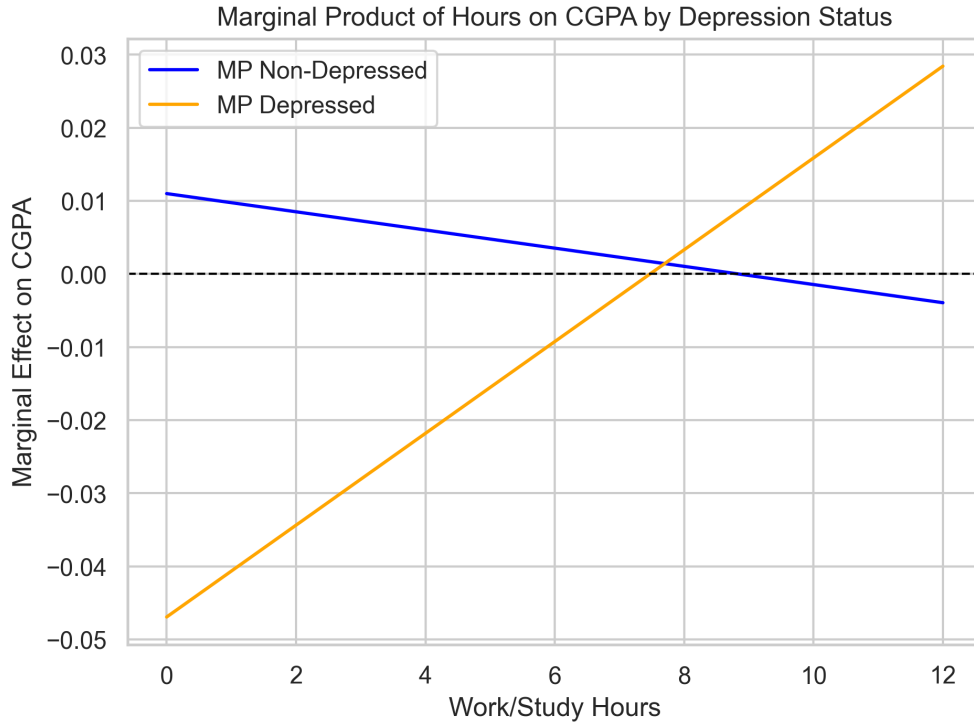


Figure 4: Marginal product of study hours for depressed and non-depressed students. The non-depressed marginal product declines with hours, whereas the depressed marginal product starts lower but increases with hours in the estimated quadratic specification.

Thus, depressed students operate at roughly 65% of the hourly productivity of non-

depressed students. This reduction in marginal product plays a central role in the dynamic programming analysis, as it shifts the trade-off between effort and GPA gains and alters the optimal study-time policy.

6 Economic Interpretation

The marginal product calculations indicate that an additional hour of study raises GPA only slightly for both groups, but the gain is systematically lower for depressed students. At their respective average study levels, non-depressed students achieve a marginal product of 0.00324, while depressed students achieve only 0.00211. In economic terms, depression shifts the productivity schedule downward: each additional hour of effort yields smaller returns. This implies that depressed students face a steeper trade-off between effort and performance, since they must exert more hours to achieve the same increment in GPA. These diminished returns help explain the higher study hours observed among depressed students and motivate the dynamic programming analysis of optimal effort choice under differing productivity profiles

7 Utility Maximization Framework

To connect the empirical production estimates to an economic theory of student effort, we model each student as choosing daily study hours H to maximize a per-period payoff that values academic progress but penalizes study effort. Let G denote current GPA. Next-period GPA evolves according to

$$G' = G + f(H; d),$$

where $d \in \{0, 1\}$ indexes depression status and $f(H; d)$ is the estimated GPA increment from studying H hours. The function $f(H; d)$ differs across variants of the empirical specification, and the dynamic model incorporates each of these forms.

Students obtain utility from higher GPA but incur a disutility from studying. In the implementation, the payoff from GPA is assumed to be *concave*, reflecting diminishing marginal utility of academic achievement. Specifically, letting $u(\cdot)$ denote the GPA utility function, the instantaneous payoff is

$$U(G, H; d) = u(G') - \chi_d H,$$

where $\chi_d > 0$ is the marginal disutility of effort for type d . Consistent with the code, $u(\cdot)$ follows the CRRA family:

$$u(G) = \begin{cases} \log(G), & \text{if } \sigma = 1, \\ \frac{G^{1-\sigma} - 1}{1-\sigma}, & \text{if } \sigma \neq 1, \end{cases}$$

with $\sigma = 1$ (log utility) used as the baseline specification.

In dynamic form, students choose H to maximize discounted lifetime utility:

$$V(G; d) = \max_{H \in [0, 12]} \{u(G') - \chi_d H + \beta V(G'; d)\}, \quad 0 < \beta < 1.$$

The code implements this dynamic problem by discretizing the GPA state space and solving for the optimal policy $H^*(G; d)$ using value iteration. Because the empirical production function admits several parametric forms, we evaluate optimal study policies under multiple transition specifications $f(H; d)$.

8 Dynamic Programming Model

We embed the empirical productivity differences into a simple dynamic programming (DP) setting. The state variable is GPA G and the control is study time h . The code computes four alternative versions of the transition function, reflecting different assumptions about how GPA responds to hours.

8.1 Empirical Transition Structures

Let G_t denote GPA at time t . The model computes optimal study hours under three specifications for the GPA increment $f(H; d)$:

1. **Linearized productivity (LIN):**

$$f_{\text{lin}}(H; d) = MP_d(\bar{H}) \cdot H,$$

where $MP_d(\bar{H})$ is the empirical marginal product for type d evaluated at mean study hours.

2. **Quadratic full (QF):**

$$f_{\text{qf}}(H; d) = (b_1 + g_2 d)H + (b_2 + g_3 d)H^2,$$

where (b_1, b_2, g_2, g_3) are estimated from the empirical regression.

3. **Centered regression (CC):**

$$f_{\text{cc}}(H; d) = (b_1^c + g_2^c d)H_c + (b_2^c + g_3^c d)H_c^2, \quad H_c = H - \bar{H},$$

where coefficients $(b_1^c, b_2^c, g_2^c, g_3^c)$ come from the centered polynomial regression estimated in the code. This variant typically produces the most stable marginal product measures.

Each specification yields a different state transition

$$G_{t+1} = G_t + f_\ell(H_t; d),$$

with $\ell \in \{\text{lin}, \text{qf}, \text{cc}\}$.

8.2 Instantaneous Utility

In all specifications we treat effort cost additively but allow the student's payoff from GPA to be *concave*. Denote by $u(\cdot)$ the per-period utility derived from next-period GPA G_{t+1} . The instantaneous payoff is therefore

$$u(G_{t+1}) - \chi_d h_t.$$

In the code we implement the standard CRRA family for $u(\cdot)$, with parameter σ :

$$u(G) = \begin{cases} \log(G) & \text{if } \sigma = 1, \\ \frac{G^{1-\sigma} - 1}{1-\sigma} & \text{if } \sigma \neq 1, \end{cases}$$

and we impose a small numerical floor to ensure $G > 0$ inside the utility function. The linear disutility of effort remains $\chi_d h_t$, where χ_d is calibrated from the empirical marginal product at mean hours as in the code:

$$\chi_0 = MP_{\text{non}}(\bar{H}), \quad \chi_1 = MP_{\text{dep}}(\bar{H}).$$

In all numerical results presented below, we set $\sigma = 1$ so that $u(G) = \log(G)$, following the instructor's recommendation for a smooth and concave specification.

9 Policy Functions Across Specifications

The DP solver produces 3 pairs of policy functions:

$$\{h_{\text{lin},d}^*(G), h_{\text{qf},d}^*(G), h_{\text{cc},d}^*(G)\}, \quad d \in \{0, 1\}.$$

A key result from the code is that *the relationship between depression and optimal hours is*

specification-dependent. In some cases, such as the full quadratic model, depressed students choose lower hours at certain GPA levels due to the curvature of the estimated production function. In others—most notably the centered polynomial and linear variants—the depressed policy lies above the non-depressed policy at many GPA values. Because the GPA payoff is concave in the implemented model (log/CRRA utility), the resulting policy functions are smoother than under a linear-utility formulation. Concavity reduces the prevalence of corner solutions—situations in which students study either the maximum or minimum feasible hours—and instead yields gradually declining optimal hours as GPA rises. This aligns with economic intuition: when the marginal utility of GPA diminishes at higher levels, additional study effort becomes less valuable, leading to interior optima.

Evaluating the policies at the sample mean GPA ($\bar{G} \approx 7.656$) yields, for example:

$$h_{\text{lin},0}^*(\bar{G}) \approx 8.75, \quad h_{\text{lin},1}^*(\bar{G}) \approx 12.0,$$

but the quadratic specifications may produce smaller or even reversed differences. With concave utility, these reversals are less extreme than under linear utility, but the qualitative ranking can still vary across specifications because the productivity functions imply different returns to effort across the GPA range. This reflects the fact that each empirical specification implies a different marginal productivity schedule.

Across all specifications, however, two qualitative observations hold:

- Policies tend to be weakly decreasing in GPA: students with higher GPA need fewer additional study hours to sustain performance.
- Depression affects optimal hours through two competing channels: reduced productivity (f_ℓ smaller for $d = 1$) and reduced effort cost ($\chi_1 < \chi_0$). The balance of these forces determines the ranking of $h^*(G; 0)$ and $h^*(G; 1)$, yielding specification-dependent results.

Figure 5 and 6 plots one representative set of the computed policies.

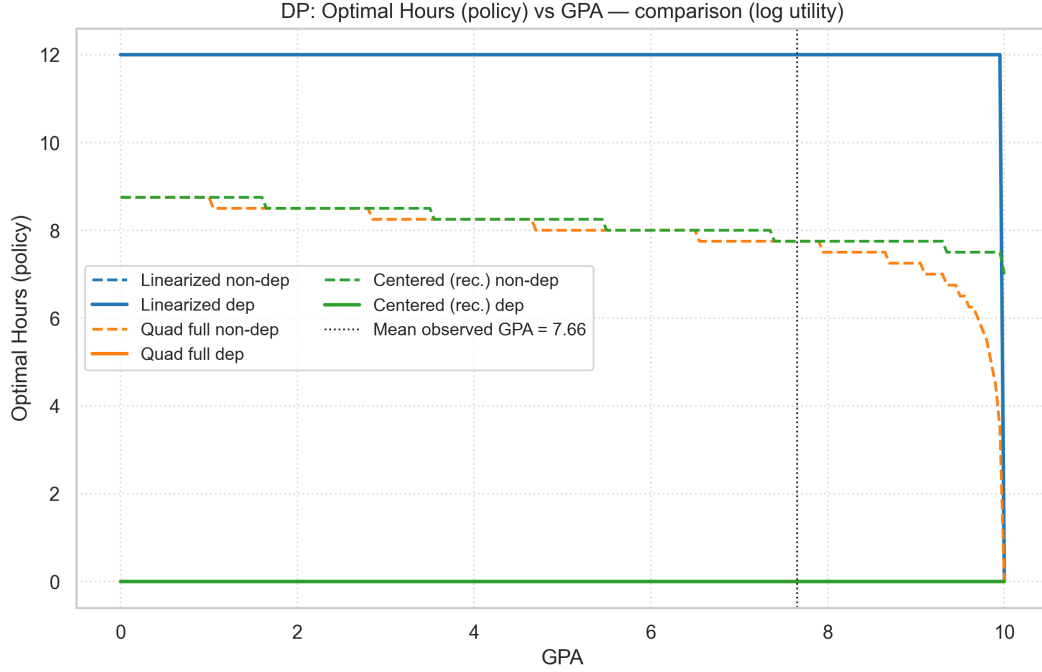


Figure 5: Comparison of optimal study-hour policies under three empirical transition specifications (linearized, quadratic full, centered). Lines compare non-depressed and depressed students; the ranking of effort is specification-dependent.

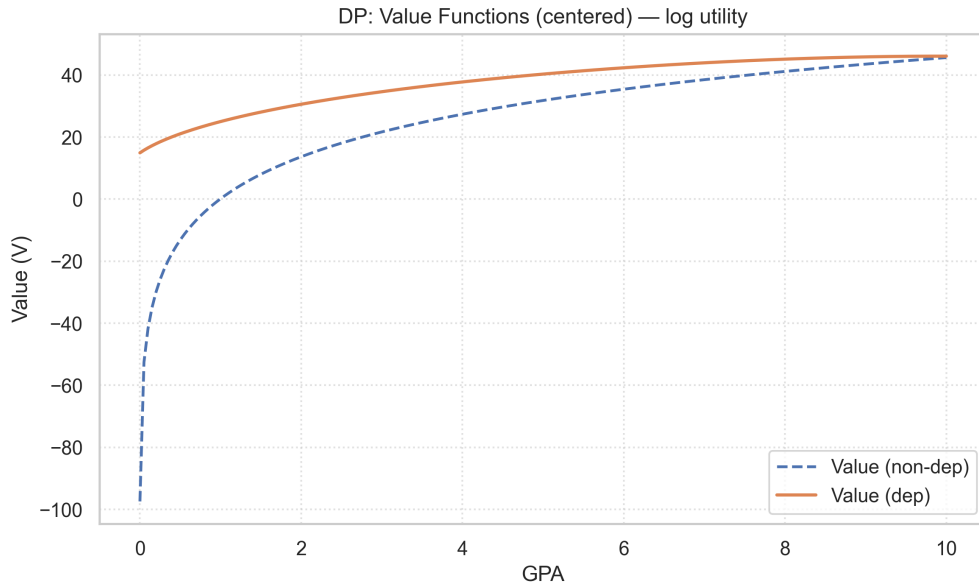


Figure 6: Value functions under the centered transition specification with log utility. The concave utility causes smooth value curves and reduces corner solutions.

10 Conclusion

The dynamic programming analysis demonstrates how different empirical assumptions about academic productivity translate into different optimal study behaviors. Because depression reduces estimated marginal productivity, optimal study hours may increase or decrease depending on how strongly the cost parameter χ_d adjusts in the calibration and how the production function is modeled.

The code’s multiple specifications show that predictions are *sensitive to the chosen functional form*. In particular, while some specifications predict higher optimal hours for depressed students, others predict smaller or reversed gaps. This highlights the importance of carefully modeling the productivity–effort relationship when using dynamic models to interpret behavioral responses to mental health conditions.